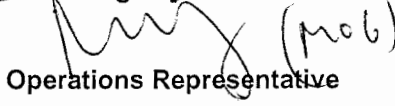


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Excore NIS		
TASK:	Perform a QPTR		
TASK NUMBER:	N0150020201		
JPM NUMBER:	13-01 NRC RO Admin A1-2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.43
APPLICABILITY:	IMPORTANCE FACTOR: <u>4.1</u>		
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S1.OP-ST.NIS-0002 Rev. 15 S1.RE-RA.ZZ-0011 Rev. 383 Tech Spec 3.2.4		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	<u>30 min</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	09-24-14
Validated By:	E Bak SME or Instructor	Date:	10-14-14
Approved By:	 Training Department	Date:	10-23-14
Approved By:	 Operations Representative	Date:	10-23-14
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE: _____ DATE: _____			

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Excore NIS

TASK: Perform a QPTR

TASK NUMBER: N0150020201

INITIAL CONDITIONS:

Unit 1 was operating at 100% power when rod 2D4 dropped fully into the core. OHA E-46, LOWER SECT DEV ABV 50% PWR annunciated and remains locked in.

Operators have not yet started reducing power to 74% to comply with TSAS 3.1.3.1.c.3.d IAW S1.OP-AB.ROD-002, Dropped Rod.

INITIATING CUE:

The CRS has directed you to perform a QPTR using Manual Calculation IAW S1.OP-ST.NIS-0002. NI currents are:

	<u>Upper Detectors</u>	<u>Lower Detectors</u>
N41	190	200
N42	200	215
N43	180	175
N44	200	205

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Perform the QPTR and calculate the highest QPTR as SAT.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Excore NIS

TASK: Perform a QPTR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide blank copy of S1.OP-ST.NIS-0002, Rev.15, Power Distribution - Quadrant Power Tilt Ratio, a copy of S1.RE-RA.ZZ-0011, Rev. 383, Tables, and a copy of Tech Spec 3.2.4.			
	2.1	IDENTIFY sections of this procedure that are <u>NOT</u> to be performed with "N/A".	N/A's Section 5.2, Attachment 1 Section 3.0, and Attachment 3.		
	3.0	<u>PRECAUTIONS AND LIMITATIONS</u>	Reads and initials PRECAUTIONS AND LIMITATIONS 3.1-3.5		
	5.1.1	IF one PR channel is inoperable, <u>AND</u> RTP is >75%, THEN...	Determines all PR channel are operable and step is NA.		
	5.1.2	RECORD the following data on Attachment 2 • Date • Time • Reactor Power • Reason for performing QPTR Calculation	Records current date, current time, 100% reactor power and checks OHA E-46 as reason for performance in Attachment 2.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Excore NIS
TASK: Perform a QPTR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.3	RECORD the following data on Attachment 1: • NI Channels N-41, N-42, N43 and N-44 Upper Detector current readings • NI Channels N-41, N-42, N43 and N-44 Lower Detector current readings • Respective 100% NI Current Values for Channels N-41, N-42, N43 and N-44 Detectors from S1.RE-RA.ZZ-0011, Table 2	Records on Attachment 1: (From initial conditions) • NI channels N41-44 Upper Detector Current Readings • NI channels N41-44 Lower Detector Current Readings • 100% NI Current Values from S1.RE-RA.ZZ-0011, TABLES NOTE: Attachment 1, Section 3 is NOT required to be performed to determine detector currents. It was added at Rev. 12 to use "when any NIS meter is suspect." If asked, CUE that all Power Range Detectors are Operable.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Excore NIS

TASK: Perform a QPTR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.1.4	COMPLETE Attachment 1 calculations.	For Top and Bottom Detectors: (numbers as per key for evaluator) - Calculates Detector Ratio for each top and bottom detector. - Adds detector ratios to get Sum of detector ratios. - Divides Sum by number of operable detectors (4) to get Average Detector Ratios. - Places Average Detector Ratios in 4th column of Att. 1 - Divides each detector ratio by the average ratio to get the power tilt for each detector. Cue: IV is complete when asked for IV of calculations		
*	5.1.5	RECORD the following on Attachment 2 "Power Tilt" for each detector. "Maximum Power Tilt" and applicable detector identification information. - Test Results by initialing SAT or UNSAT column IAW stated Acceptance Criteria.	Records information on Attachment 2 (as per key for evaluator) Maximum Power Tilt for Top and Bottom will be <1.02 and marked SAT		
	5.1.6	DIRECT a second Operator to perform Independent Verification of calculations in Attachment 1, Sections 1.0, 2.0 and 3.0 as applicable.	Cue: IV is complete SAT.		

OPERATOR TRAINING PROGRAM

NAME: _____

JOB PERFORMANCE MEASURE

DATE: _____

SYSTEM: Excore NIS

TASK: Perform a QPTR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.1.7	<u>IF</u> the Maximum Power Tilt for <u>any</u> detector exceeds 1.02, <u>THEN REFER</u> to T/S 3.2.4 for corrective actions.	Determines no tilt exceeds 1.02.		
	5.3.1	This surveillance is satisfactory when Attachment 2 or 3 is completed with the Test Data meeting the Acceptance Criteria stated.	Determines surveillance is SAT.		
	5.4.1	COMPLETE Attachment 4, Sections 1.0 and 2.0, AND FORWARD completed procedure to SM/CRS for review.			
			JPM is terminated when candidate returns all JPM paperwork to evaluator.		

OPERATOR TRAINING PROGRAM JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- SB 1. Task description and number, JPM description and number are identified.
- SB 2. Knowledge and Abilities (K/A) references are included.
- SB 3. Performance location specified. (in-plant, control room, or simulator)
- SB 4. Initial setup conditions are identified.
- SB 5. Initiating and terminating Cues are properly identified.
- SB 6. Task standards identified and verified by SME review.
- SB 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- SB 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. various Date 10-14-14
- SB 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: Ed Bahj

SME/Instructor: _____

SME/Instructor: _____

Date: 10-14-14

Date: _____

Date: _____

INITIAL CONDITIONS:

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